Rewriting the References Helped Pose-Based Kazakh Sign Language Translation More Than Any Model Change

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Abstract

We translate Kazakh Sign Language news interpretation into Kazakh text from pose keypoints alone, with a Uni-Sign pose encoder feeding mT5-base, trained on 59k clips from three broadcasters. Across 42 from-scratch runs in twelve configurations, each with its readings fixed before launch, three findings stand. First, training is bimodal. A run either starts reading its pose input ("takes off", chrF≈30) or falls back on the decoder's language prior (chrF≈20), and which one happens is largely chance. Seven of the 36 runs on the original data took off; seeds of one configuration span 10 chrF; and no change to the encoder objective, learning rates, decoder initialisation, frame rate or preprocessing moved the rate. Second, the same checkpoints that almost never write the right sentence (WER 0.96) pick it out of 200 same-source candidates 81% of the time on the test set, once the text prior is subtracted with pointwise mutual information. Third, the largest gain came from the data. Used as an auditor, retrieval shows that 13% of one broadcaster's training references describe a different part of the broadcast, an artefact of fuzzy-matching article text to time segments. Removing them changes nothing. Rewriting the references with an LLM so that they cover exactly what was spoken, in the article's wording, does two things. Without retraining, it raises that broadcaster's retrieval from 0.67 to 0.87, so about two thirds of its apparent deficit was evaluation error. Training from scratch on the rewritten references then raises test chrF from 30.3 to 35.8 (+5.5, 95% CI [4.9, 6.1]), with gains in every source. The best model gets the story but almost never the rare words (2.6% recall), and the signers' hands are 31–51 pixels across.

1 Introduction

Sign language translation (SLT) from video is usually reported as one number on one benchmark (Camgoz et al., 2018, 2020), and recent gloss-free systems put a pretrained visual or pose encoder in front of a pretrained language model (Zhou et al., 2023; Wong et al., 2024; Gong et al., 2024; Li et al., 2025). That recipe hides a failure the NLP literature has long known under other names. A decoder strong enough to write fluent text can learn to ignore its input (Bowman et al., 2016; Voita et al., 2021) and produce hallucinated, prior-driven output (Lee et al., 2018). With a weak input signal, low-resource data and noisy references, one training run cannot show which of these is happening.

This paper is a record of what did and did not improve a keypoints-only translator for Kazakh Sign Language (KSL) news interpretation, built on Uni-Sign (Li et al., 2025) and mT5 (Xue et al., 2021). We started by asking which input representation works best. Running that comparison changed the question. One configuration's three seeds spanned 19.8–29.8 chrF, wider than any difference between configurations, and the spread came from a single binary event: whether the decoder ever starts to read the encoder. Every intervention we then tried on the model side left that event's rate where it was. The changes that mattered came from treating the model as a measuring instrument for the data rather than the other way round.

Contributions. (1) Evidence, over 42 from-scratch runs, that gloss-free pose-to-text training is bimodal and that the outcome of a single run is mostly chance. Five families of model-side intervention leave the takeoff rate unchanged (§4.1). (2) A pose-shuffle measure of encoder use with per-run confidence intervals. It shows coupling is graded below the takeoff threshold, which a binary rule hides (§4.2). (3) A demonstration that the

same checkpoints solve sentence retrieval at 81–86% top-1 out of 200 while generation stays near WER 0.96, with controls for duration, memorisation and coupling (§4.3). (4) A retrieval-based corpus audit that finds the misaligned references, and an LLM rewrite of them. The rewrite exposes most of one source's deficit as evaluation error and, used for training, gives the only large gain in the study (§4.4). (5) A list of pipeline faults that each would have reversed a conclusion without raising an error (Appendix A).

2 Related Work

Gloss-free SLT. Early neural SLT used gloss supervision (Camgoz et al., 2018, 2020), which is costly to annotate and a lossy description of signing (Müller et al., 2023). Gloss-free systems instead align visual features with text through contrastive pretraining (Zhou et al., 2023) or connect a visual encoder to a frozen or tuned LLM (Wong et al., 2024; Gong et al., 2024). Uni-Sign (Li et al., 2025) pretrains a pose and RGB encoder on 1,985 hours of Chinese news interpretation (CSL-News) and fine-tunes it with mT5 for translation. It is our encoder and our template. Large web corpora such as YouTube-ASL (Uthus et al., 2023) and BOBSL (Albanie et al., 2021), and benchmarks such as How2Sign (Duarte et al., 2021) and CSL-Daily (Zhou et al., 2021), show the same broadcast-interpretation setting and its noise.

Kazakh–Russian Sign Language. Prior KRSL work covers isolated-sign and linguistic datasets (Imashev et al., 2020; Mukushev et al., 2022). To our knowledge there is no published continuous KSL-to-Kazakh translation result to compare against. The three-broadcaster corpus used here was collected and segmented at ISSAI and has not been published separately.

Decoders that ignore the encoder. Posterior collapse in VAEs (Bowman et al., 2016), analyses of source versus target contributions in NMT (Voita et al., 2021) and NMT hallucination (Lee et al., 2018) all describe a conditional decoder that relies on its own prefix. We measure the sign-language case directly by permuting encoder outputs. To counter the prior at scoring time we use the mutual-information correction from dialogue (Li et al., 2016) and zero-shot scoring (Holtzman et al., 2021).

Noisy and misaligned references. Noise in parallel data hurts NMT unevenly depending on its type (Khayrallah and Koehn, 2018). In sign language corpora built from broadcast subtitles, the subtitles are offset from the signing, and Bull et al. (2021) learn to re-align them in time. Our noise is different: the text, not its timing, belongs to another part of the broadcast. Rewriting captions with a language model has improved vision–language training (Fan et al., 2023). We apply the same idea to sign language references and measure its effect on both training and evaluation. Duarte et al. (2022) study sign video retrieval from free-form text queries; we use retrieval both as a task and as a data-audit tool.

Seed variance. Fine-tuning outcomes can depend on the seed as much as on the method (Dodge et al., 2020; Bouthillier et al., 2021). Our bimodal takeoff is an extreme case, and it forces the design used throughout: several seeds per arm, readings fixed in advance, and continuous measures where the binary outcome lacks power.

3 Setup

3.1 Data

The corpus holds sign language interpretation insets from three Kazakh news broadcasters, segmented into sentence-level clips with Kazakh transcripts and pre-extracted COCO-WholeBody keypoints (Jin et al., 2020) from RTMW (Jiang et al., 2024). It totals about 360 hours. The three sources differ in ways that matter later (Table 1). INFORMBURO and KHABAR carry one caption-style transcript per clip. QAZAQSTANTV stores two: `text_asr`, a Whisper transcription (Radford et al., 2023) of exactly what the anchor said in the clip's window, and `text_human`, sentences of the broadcast's web article fuzzy-matched onto time segments. Training used `text_human` where a match existed (89.8%) and `text_asr` otherwise.

We rebuilt the splits because the released versions leaked: an archived QAZAQSTANTV training split shared 563 videos with the recollected dev split, 64% of dev clips. Canonical splits are assigned per source video by a salted hash: 59,008 train clips (10,398 videos), 8,411 dev (2,575) and 8,360 test (2,534). All model selection uses a frozen *selection set* of 600 dev clips: 200 per source, one per video, stratified over duration quar-

Source	fps	% train	hand px	reference
INFORMBURO	50	9	34	caption
KHABAR	25	36	51	caption
QAZAQSTANTV	30	55	31	article / ASR

Table 1: The three sources. Hand size is the median extent of the 21-point hand box over 40 dev clips per source. Videos are interpreter crops of 204×186 to 314×282 pixels, and no higher-resolution originals are stored.

tiles. Test was opened twice, each time for one checkpoint chosen on dev: once for retrieval (§4.3) and once for the final model (§4.4).

3.2 Model

The encoder is Uni-Sign's pose network (Li et al., 2025), initialised from its CSL-News stage-1 weights. It uses 69 keypoints per frame, split into body (9), face (18) and each hand (21), each with (x, y, score) and preprocessed exactly as upstream: a clip-level body box, wrist-relative hands and nose-relative face. Per part, a linear layer feeds three spatial ST-GCN blocks (Yan et al., 2018) (64–128–256 channels); body features are added into the hand and face branches; three temporal blocks (kernel 5) follow; joints are mean-pooled. The four parts are concatenated and projected to 768 dimensions (5.3M parameters in total). A LayerNorm brings the features to mT5's embedding scale; without it they sit about six times larger and nearly parallel across clips. The normalised frames are appended to the embedded prompt "Translate sign language video to Kazakh:" and passed as `inputs_embeds` to mT5-base (Xue et al., 2021) (580M). The whole model is trained end to end with cross-entropy. Padding is re-zeroed after every temporal block so that convolution bias cannot leak into real frames.

3.3 Training and decoding

Unless stated otherwise every run uses the same recipe. Clips are taken at every second frame (25/12.5/15 fps for INFORMBURO/KHABAR/QAZAQSTANTV; §4.1 tests alternatives). Training runs 10 epochs with AdamW, batch 8×4 accumulation and cosine decay; the learning rate is 5×10^{-5} for the encoder and 5×10^{-4} for mT5 and the LayerNorm. Decoding uses beam 4, repetition penalty 1.3 and trigram blocking. A decoding sweep moved chrF by at most 0.27 within beam 4 and showed beam 4 ahead of greedy by 1.3 (Appendix D). Each run takes about 5.7 hours on one RTX PRO 6000 GPU.

3.4 Measures

Generation. We report corpus chrF (Popović, 2015) via sacreBLEU (Post, 2018), and *content recall*: the share of reference words of at least four characters that appear in the hypothesis. BLEU and WER are near floor for every model (WER 0.92–1.04) and are given in the appendix only.

Takeoff. A run *takes off* if its epoch-10 checkpoint reaches chrF ≥ 26 and content recall ≥ 0.08 on the selection set. The thresholds were fixed from the first six runs, before any later arm was scored; the two modes are far apart (Figure 1).

Coupling. To measure how much a checkpoint reads its encoder, we decode the selection set twice. The second pass permutes the encoder output across the batch with a fixed derangement, moving the pose mask with it, so that each clip is decoded from another clip's pose. Then

$$\Delta_{\text{pose}} = \frac{1}{N} \sum_{i} \left[\text{chrF}(\hat{y}_i) - \text{chrF}(\hat{y}_i^{\text{shuf}}) \right], \quad (1)$$

with sentence-level chrF, and a 95% bootstrap interval over clips. A decoder that ignores its input has $\Delta_{\text{pose}} \approx 0$.

Retrieval. We score candidate sentences t for a clip x by the checkpoint's own teacher-forced likelihood and correct for each sentence's prior probability:

$$s(t, x) = \log p(t \mid x) - \frac{1}{3} \sum_{k=1}^{3} \log p(t \mid x_k), \quad (2)$$

where x_k are poses from other videos of the same source. Candidates always come from the query's own source, so that differences in transcript style (QAZAQSTANTV is cased and punctuated) cannot be used. **T1** is 200-way search: the clip's sentence among those of 199 other videos. **T2** is within-broadcast alignment: the clip's sentence among all sentences of its own broadcast (broadcasts with at least 5 clips, mean pool 6.4), where every candidate shares topic and names.

Protocol. Before each experiment we wrote its design, primary measure and a table of outcomes

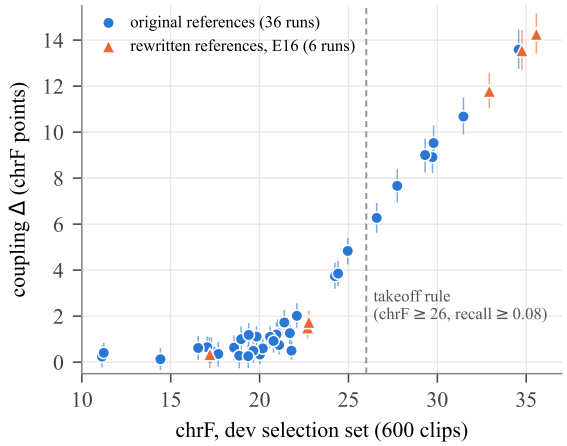


Figure 1: All 42 from-scratch runs at epoch 10 on the dev selection set. Error bars are 95% bootstrap intervals over clips. Runs fall into two groups with nothing between chrF 25 and 26.5. Coupling rises with chrF ($r = 0.93$) but separates runs the binary rule cannot, for example partial runs at Δ_{pose} 3.7–4.8. The triangles are E16, trained on rewritten references (§4.4).

with their readings into the experiment log, and we report the reading that fired, including where it later proved too strong. Where an experiment compares fine-tuned arms we include two identical arms with different seeds, which gives a measured noise floor. Takeoff rates are compared with Fisher's exact test; paired chrF differences use paired bootstrap resampling (Koehn, 2004).

4 Results

4.1 Training is bimodal, and the model-side levers do not move it

One configuration, three outcomes. Our first experiment (E3) compared five input pipelines, one seed each. Upstream Uni-Sign preprocessing led at 29.8 chrF against 18.3–22.9 for the others, and the obvious reading was that compatibility with the pretrained weights matters. Three seeds of the winning configuration then scored 29.8, 21.0 and 19.8. Every other arm fell inside that spread, so we withdrew all five comparisons. The training curves show why. All runs follow one path through epoch 4; the run that succeeds leaves it at epoch 5 and ends at a dev cross-entropy of 2.29, while the others flatten at 2.69–2.75 and cycle through a few stock broadcast phrases. In the worst case, one sentence ("now let us turn to the main news that shook the world") is emitted for 140 of 600 clips; it is the most repeated reference in the training set. "Seed" here labels a run rather than reproducing it: under the default non-deterministic kernels, two runs with seed 0 reached losses of 18.34 and 17.26 by batch 400.

Arm	Change	Takeoff	Takeoff chrF	Δ_{pose} fail
B	baseline	1/3	29.8	1.19
E4	mT5 LR ÷10	0/3†	–	0.52
E4b	encoder LR ×2	2/6	29.7, 31.5	1.29
E5	seq. alignment	0/3	–	1.49
E6	token alignment	1/3	29.3	1.41
E7	adapted mT5	0/1	–	0.28
E9	12.5 fps	0/2	–	0.55
E9b	15 fps	2/6	34.6, 27.7	0.90
E10	native fps	0/3	–	0.30
E11	native, cap 256	1/3	26.6	0.63
E13	drop flagged refs	0/3	–	2.98
all, original refs		7/36	26.6–34.6	
E16	rewritten refs	3/6	35.6/34.8/32.9	1.20

Table 2: From-scratch arms (dev selection set, epoch 10). Δ_{pose} fail is the mean coupling of runs that did not take off. †E4 is undertrained (train CE 3.33–3.36 against the baseline's 2.12–2.50) and cannot be classified. Per-run numbers are in Appendix C.

What we varied. Table 2 lists every from-scratch arm, each judged against the three baseline seeds (B). The arms covered each place a lever could plausibly sit:

- *encoder objective*: InfoNCE (van den Oord et al., 2018) between pooled pose embeddings and a frozen mT5 text embedding (E5), and a token-level version (E6);
- *optimisation*: mT5 slower than the encoder (E4), and the encoder twice as fast (E4b);
- *decoder initialisation*: mT5 domain-adapted by span corruption (Raffel et al., 2020) on our training transcripts, cutting held-out perplexity from 15.5 to 4.6 (E7);
- *temporal input*: all sources resampled to a common 12.5 or 15 fps (E9, E9b), native frame rate (E10), and native rate capped Uni-Sign-style at 256 randomly sampled frames (E11);
- *data*: the 4,200 audit-flagged QAZAQSTANTV references removed (E13; §4.4).

None of them moved the rate. Across the 36 runs on the original references, 7 took off (19%). No arm differs from baseline in takeoff rate, and at these sample sizes none could have shown a moderate change. More useful is what each arm did to the quantity it was meant to change. Both alignment arms drove their contrastive loss from chance (5.58) to 1.2–2.0: the encoder learned to identify

its own transcript, and the decoder still did not use it. The adapted decoder fitted the text faster (train CE 4.1 against 5.0 after one epoch) and then collapsed onto the same stock phrase as the failed baselines. Twice we reported a ceiling or a rate change from a small sample, and twice more seeds removed it. E4b's 2/3 became 2/6. E9b's first takeoff reached 34.6 chrF and coupling 13.6, outside every earlier run, but its next takeoff landed at 27.7 and 7.7, below every earlier run. We keep both corrections in the record because they are the reason the protocol above exists.

The frame-rate series. Clips were taken at every second frame on the assumption that all videos were 50 fps; only INFORMBURO is. Correcting this made results worse. At native rate (E10) every coupling interval includes zero and one seed emits one sentence for 60% of clips. Uni-Sign never fed its encoder more than 256 frames, and capping native-rate clips at 256 random frames (E11) recovers about half the loss. Lowering the rate everywhere (E9) also hurt. Stride 2, which keeps clips at 220–390 frames, remains the best setting we have tested. The likeliest reading is that the pretrained encoder works best near the sequence statistics it was trained on. That reading was formed after the fact and is untested.

4.2 Coupling is graded below the threshold

Takeoff is a binary outcome with a base rate near 20%, so detecting a change from one third to two thirds would take more seeds than we could afford. The coupling measure gives each run a value with its own interval. At the corpus level it adds little: over non-degenerate runs the shuffled score is nearly constant (18.5–21.1 chrF), so corpus Δ_{pose} is almost clean chrF minus a constant. That constant is informative in itself. It is the chrF of the text prior, and every run that did not take off scores at or just above what a good run scores with its encoder switched off.

Paired per clip, the measure has resolution the binary rule lacks. Eleven of the first fifteen non-takeoff runs have intervals that exclude zero, so they read their encoder a little. Alignment raised coupling in the failed runs (1.41–1.49 against 1.19), as intended, but never enough to cross the threshold. Slowing the decoder (E4) lowered it to 0.52. Partial runs sit between the groups (E4b_s5: chrF 24.2, Δ_{pose} 3.74). The measure also ranks runs the rule treats as equal: two takeoffs of the same arm, E9b, have non-overlapping intervals (13.6 vs 7.7).

Task	Source	top-1	top-5	MRR	chance
T1	Inf.	0.835	0.930	0.882	0.005
(200-way)	Kha.	0.885	0.945	0.910	0.005
	Qaz.	0.700	0.780	0.745	0.005
	all	**0.807**	0.885	0.845	0.005
T2	Inf.	0.900	–	0.939	0.162
(own	Kha.	0.965	–	0.979	0.175
broadcast)	Qaz.	0.639	–	0.749	0.147
	all	**0.825**	–	0.882	0.161

Table 3: Retrieval on the test split with B_s0 (original references), opened once. The checkpoint was chosen on dev by T2 MRR under a fixed rule: three takeoff checkpoints were within 0.008, inside a pre-set tie band, so the tie went to the default pipeline. Raw likelihood without the prior correction gives T1 top-1 0.005–0.015.

4.3 Retrieval works where generation does not

Generation asks the model to get every word right from weak, partial evidence, and it loses that contest to the text prior. Retrieval only asks it to rank, and a few correctly read signs (a name, a number, a place) can single out one sentence. Scored as retrieval, the baseline takeoff checkpoint B_s0, whose generated output has WER 0.96, ranks the right sentence first among 200 for 81% of test clips, with median rank 1 (Table 3). The signal appears only after correcting for the text prior: raw likelihood is at chance. Three controls on dev rule out shortcuts. (i) *Duration.* Ranking by length match, with a length model fitted on the evaluation clips themselves, gives top-1 0.010. (ii) *Memorisation.* The nearest training sentence shares a median 9–12% of content words with a dev reference, and 1 of 400 queries reaches 50%. (iii) *Coupling.* The non-takeoff seed B_s1 (Δ_{pose} = 1.27) reaches 0.015 top-1 where B_s0 (Δ_{pose} = 9.52) reaches 0.855, so retrieval requires a model that reads its input. A second takeoff checkpoint from another arm (E4b_s1) replicates the result at 0.850. QAZAQSTANTV is the weak source in both tasks, by 0.2–0.3. The next section shows why.

4.4 The references were the bottleneck

Retrieval as an auditor. We score every clip in a broadcast against every sentence of that broadcast

and flag a *mutual mismatch* when another sentence fits the clip better and the clip's own sentence fits another clip better. This is stricter than a single ranking error, which is often ordinary model error. Over 16,174 dev and test clips, 0.4% of KHABAR and 1.5% of INFORMBURO clips are flagged, against 11.5% of QAZAQSTANTV. On the 58,437 training clips the figures are 0.4%, 1.0% and 13.1% (4,200 clips). B_s0 retrieves its own training clips no better than unseen dev clips (QAZAQSTANTV 0.689 vs 0.670), so it has not memorised its training pairs and can audit them directly. The flags were checked against a signal the model never sees, implied speech rate. Clips whose transcript implies fewer than 1 or more than 4 words per second are flagged 9.8 times as often on train and 12.5 times on dev. The source of the problem is the article matching: within QAZAQSTANTV, clips with text_human references are flagged 2.8 times as often as clips with text_asr, and 3.9 times in long broadcasts. This holds even though the ASR clips come from larger broadcasts, which makes retrieval harder for them.

Removing the flagged references does nothing. Training from scratch without the 4,200 flagged clips (E13) gave no takeoff in three seeds and no QAZAQSTANTV-specific change in partial runs. To remove the takeoff lottery, E14 fine-tuned B_s0 for three epochs on four datasets: cleaned, minus 4,200 *random* QAZAQSTANTV clips, and two identical full copies, which give the noise floor. Removing the flagged clips and removing random ones differ by −0.003 on QAZAQSTANTV T2 top-1, against a measured threshold of 0.044, and neither differs from full data. Dropping a misaligned reference also drops its clip, and the remaining references are loose in a way the conservative flag does not catch.

Rewriting them. The two QAZAQSTANTV transcripts fail in opposite ways: the ASR has the right boundaries and noisy words, and the article has the right words and the wrong boundaries. We give Qwen3.6-27B (Qwen Team, 2026), served with vLLM (Kwon et al., 2023) using greedy decoding and thinking off, the clip's ASR and the article text of the clip and its two neighbours on each side. We ask it to write exactly the ASR's span in the article's wording (prompt in Appendix B). It never sees the current reference. A guard, fixed on a 600-clip dev pilot before the full run, accepts a rewrite only if its length is within $[0.67, 1.5]\times$ that of the ASR, with Whisper's duplicated passages collapsed, and its chrF against the ASR is at least 45. Rejected clips keep their reference. On 34,768 clips, which took 72 minutes on two GPUs, 96.5% of train and 97.0% of dev rewrites were accepted. Median chrF between rewrite and original is 93, and 7.9% fall below 30, which means nearly replaced.

Dev refs	Inf.	Kha.	Qaz.	mismatch
original	0.925	0.967	0.670	20.4%
LLM-rewritten	0.925	0.967	**0.869**	6.2%
raw ASR	0.925	0.967	0.877	6.2%

Table 4: T2 dev top-1 of a fixed model (B_s0) under three versions of the QAZAQSTANTV dev references (Inf./Kha./Qaz. = INFORMBURO/KHABAR/QAZAQSTANTV; mismatch is the QAZAQSTANTV mutual-mismatch rate). The other sources are untouched, and their scores do not move.

Most of the QAZAQSTANTV deficit was evaluation error. B_s0 never saw dev, so it can judge which version of a dev reference fits the signing (Table 4). Rewriting only QAZAQSTANTV's dev references, with the model unchanged, raises its T2 top-1 from 0.670 to 0.869 and lowers its mutual-mismatch rate from 20.4% to 6.2%. Of the 0.30 gap to KHABAR, 0.20 was in the references. This is not a length effect: rewritten references are longer on average, but the gain is also large where the rewrite is shorter (0.705 to 0.803), and it grows with how much the text changed. The 39 references that were nearly replaced go from 10% to 85% retrievable.

Training on rewritten references. Fine-tuning B_s0 on rewritten training references (E15, two seeds) raised QAZAQSTANTV chrF by 3.2 against the two identical full-data arms, which differ from each other by 0.28. INFORMBURO and KHABAR stayed within their spread, and the QAZAQSTANTV output/reference length ratio rose from 0.84 to 0.97: trained on fragments, the model had learned to stop early. The pre-registered retrieval measure moved by +0.011, under its bar of 0.044, because a coupled model is already near its retrieval ceiling. Training from scratch on the rewritten references (E16, six seeds; dev and test references left original) is the only from-scratch arm that changed the level of the good mode. Its three takeoffs reach 35.6, 34.8 and 32.9 chrF, against 26.6–34.6 for every takeoff on the original data (mean 29.9), and coupling 11.8–14.3 against 6.3–13.6. Gains ap-

pear in every source, not only the cleaned one, and QAZAQSTANTV now scores above INFORMBURO. The takeoff rate, 3/6 against 7/36, is not significantly different ($p = 0.13$).

The result holds on test. E16_s0, the best dev run, was compared on test with the baseline B_s0 (Table 5). The test references were rewritten with the frozen prompt and guard (97.3% accepted), and both versions are reported. chrF rises by 5.5 [4.9, 6.1], coupling by 4.6, and retrieval in both tasks. The test set has the same reference problem as dev: B_s0's QAZAQSTANTV T2 rises from 0.651 to 0.784 when only the references change. Every QAZAQSTANTV number scored against the original references, including those in this paper's earlier sections, understates the model.

4.5 What the best model gets wrong

At the median clip the model names the right topic, place or figure in fluent Kazakh and fills the rest with plausible, wrong detail. Table 6 makes this concrete. Common words are recovered about a third of the time and numbers about half the time, but rare words only 2.6% of the time, and 80% of the model's content words are not in the reference. Most rare words are names and places, which interpreters fingerspell or render with name signs. Here a hand spans 31–51 pixels and adjacent finger joints are 2–3 pixels apart (Table 1), so handshape is at the edge of what any keypoint detector can resolve, and better re-detection is impossible because no larger originals exist. The best source on every measure, KHABAR, is the one with the largest hands. With three sources, that is a hypothesis.

Recall also falls along the sentence. Pose-attributable recall (clean minus shuffled) drops from 0.173 in the first fifth of the reference to 0.080 in the last, a factor of 2.16. The drop is the same in every takeoff run and survives restricting to hypotheses at least as long as their reference. The missing end-of-sentence content does not reappear in the next clip, as interpreter lag would predict (the end-minus-start difference in the adjacent clip is −0.002 to +0.001, SE ≈ 0.004). The likeliest explanation is that it was never signed: simultaneous interpreters compress fast news speech. If so, a speech transcript is a ceiling no signing-based model can reach, and part of what looks like model error is reference content with no counterpart in the input. Testing this needs sign-faithful annotation. We have built the annotation tool and a 100-clip package without audio, but no interpreter has worked through it yet.

5 Conclusion

We set out to find the best model for translating Kazakh Sign Language news from pose, and most of what we learned is about measurement. Training is a lottery between reading the pose and reciting the prior, and no change we made to the encoder, the optimiser, the decoder's initialisation or the temporal input changed the odds. A paired pose-shuffle measure shows the lottery is not all-or-nothing and gives each run an interval that a takeoff count cannot. The same checkpoints that write poor sentences retrieve the right one 81–86% of the time, which makes retrieval both a usable task today and a tool for auditing the corpus. The audit found that one source's references often describe the wrong part of the broadcast. Rewriting them with an LLM showed that most of that source's deficit was in the evaluation set, and training on the rewrites gave the study's only large gain, +5.5 test chrF across all sources. The model now gets the story and misses the specifics. With hands 31–51 pixels wide and interpreters who compress what they hear, the remaining gap may lie in the input and the references as much as in the model.

For similar projects we recommend: run at least three seeds before comparing arms, and report coupling with intervals alongside any takeoff count; check every reference set with the model before trusting per-source numbers; prefer rewriting noisy references to dropping them; and treat retrieval as a deliverable whenever candidate text exists, such as the day's articles.

Limitations

One encoder, one decoder. All runs use Uni-Sign's pose encoder with mT5-base, and other architectures, larger decoders or RGB input may behave differently. RGB was outside the project's scope. **Seeds.** Most arms have three seeds, and E7 has one. Takeoff rates near 20% cannot be compared reliably at that size, so no arm is shown to leave the rate unchanged; we show only that none changed it detectably. **The rewrite.** The LLM can introduce errors, and the guard checks only length and overlap against the ASR, not meaning. The rewrites have not been checked by a native speaker

	Generation (600-clip test selection set)						Retrieval top-1			
Model	chrF	recall	Inf.	Kha.	Qaz.	Δ_{pose} [95% CI]	T1	T2	T1 Qaz.	T2 Qaz.
B_s0	30.27	0.135	27.3	34.3	28.7	9.39 [8.62, 10.17]	0.807	0.835	0.700 / 0.785	0.651 / 0.784
E16_s0	**35.76**	**0.195**	31.8	39.2	36.2	13.95 [13.10, 14.80]	**0.853**	**0.863**	0.745 / 0.885	0.688 / 0.826

Table 5: Test split, original references unless marked. Both checkpoints were fixed on dev before test was opened. Paired bootstrap chrF difference: +5.49 [4.90, 6.08]. For the two QAZAQSTANTV retrieval columns, the values are original / rewritten test references. On rewritten references E16_s0 reaches chrF 36.18, T1 0.900 and T2 0.912.

Reference word band	*n*	B_s0	E16_s0
common (≥500 in train)	4,603	0.237	0.323
mid (20–500)	7,186	0.094	0.159
rare (<20)	2,325	0.006	**0.026**
unseen in train	530	0.000	0.000
numbers	477	0.382	0.532
content-word precision		0.148	0.201

Table 6: Recall of reference words by training frequency, dev selection set. Precision is the share of the hypothesis's content words that appear in the reference.

against the signing. Retrieval on rewritten references improves with how much the text changed, which supports the rewrite but is also measured by a model trained on the old references. **References are speech, not signing.** Every reference transcribes what the anchor said, and §4.5 suggests interpreters omit part of it, so all absolute scores mix model error with untranslatable reference content. **One test set, opened twice.** Each opening used one checkpoint fixed on dev, but the same test clips served both. **Comparability.** chrF is sacreBLEU's default chrF2 on raw text, so it is case- and punctuation-sensitive, which affects QAZAQSTANTV (cased and punctuated) more than the other sources. No prior continuous KSL translation result exists to compare against.

Ethics Statement

The data are broadcast interpretation held under a restricted-access agreement and are not redistributed. Interpreters appear in the source videos, and only keypoints were used for training. Output of the quality reported here should not be shown to Deaf viewers as a translation: it invents specific names and places 80% of the time. Retrieval against known text is the safer use. Deaf signers and interpreters have not yet taken part in evaluating the system. A 100-clip study in which an interpreter marks what was actually signed is prepared but not yet carried out (§4.5), and involving the Deaf community in evaluation is the next step of this work.

References

Samuel Albanie, Gül Varol, Liliane Momeni, Hannah Bull, Triantafyllos Afouras, Himel Chowdhury, Neil Fox, Bencie Woll, Rob Cooper, Andrew McParland, and Andrew Zisserman. BBC-Oxford British Sign Language dataset. *arXiv preprint arXiv:2111.03635*, 2021.

Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Naz Sepah, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Dmitriy Serdyuk, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems (MLSys)*, 2021.

Samuel R. Bowman, Luke Vilnis, Oriol Vinyals, Andrew Dai, Rafal Jozefowicz, and Samy Bengio. Generating sentences from a continuous space. In *Proceedings of the 20th SIGNLL Conference on Computational Natural Language Learning (CoNLL)*, 2016.

Hannah Bull, Triantafyllos Afouras, Gül Varol, Samuel Albanie, Liliane Momeni, and Andrew Zisserman. Aligning subtitles in sign language videos. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

Necati Cihan Camgoz, Simon Hadfield, Oscar Koller, Hermann Ney, and Richard Bowden. Neural sign language translation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

Necati Cihan Camgoz, Oscar Koller, Simon Hadfield, and Richard Bowden. Sign language transformers: Joint end-to-end sign language recognition and translation. In *Proceedings of the*

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah Smith. Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping. *arXiv preprint arXiv:2002.06305*, 2020.

Amanda Duarte, Shruti Palaskar, Lucas Ventura, Deepti Ghadiyaram, Kenneth DeHaan, Florian Metze, Jordi Torres, and Xavier Giro-i Nieto. How2Sign: A large-scale multimodal dataset for continuous American Sign Language. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

Amanda Duarte, Samuel Albanie, Xavier Giró-i Nieto, and Gül Varol. Sign language video retrieval with free-form textual queries. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

Lijie Fan, Dilip Krishnan, Phillip Isola, Dina Katabi, and Yonglong Tian. Improving CLIP training with language rewrites. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Jia Gong, Lin Geng Foo, Yixuan He, Hossein Rahmani, and Jun Liu. LLMs are good sign language translators. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

Ari Holtzman, Peter West, Vered Shwartz, Yejin Choi, and Luke Zettlemoyer. Surface form competition: Why the highest probability answer isn't always right. In *Proceedings of EMNLP*, 2021.

Alfarabi Imashev, Medet Mukushev, Vadim Kimmelman, and Anara Sandygulova. A dataset for linguistic understanding, visual evaluation, and recognition of sign languages: The K-RSL. In *Proceedings of the 24th Conference on Computational Natural Language Learning (CoNLL)*, 2020.

Tao Jiang, Xinchen Xie, and Yining Li. RTMW: Real-time multi-person 2D and 3D whole-body pose estimation. *arXiv preprint arXiv:2407.08634*, 2024.

Sheng Jin, Lumin Xu, Jin Xu, Can Wang, Wentao Liu, Chen Qian, Wanli Ouyang, and Ping Luo. Whole-body human pose estimation in the wild. In *European Conference on Computer Vision (ECCV)*, 2020.

Huda Khayrallah and Philipp Koehn. On the impact of various types of noise on neural machine translation. In *Proceedings of the 2nd Workshop on Neural Machine Translation and Generation*, 2018.

Philipp Koehn. Statistical significance tests for machine translation evaluation. In *Proceedings of EMNLP*, 2004.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.

Katherine Lee, Orhan Firat, Ashish Agarwal, Clara Fannjiang, and David Sussillo. Hallucinations in neural machine translation, 2018. OpenReview, ICLR 2019 submission.

Jiwei Li, Michel Galley, Chris Brockett, Jianfeng Gao, and Bill Dolan. A diversity-promoting objective function for neural conversation models. In *Proceedings of NAACL-HLT*, 2016.

Zecheng Li, Wengang Zhou, Weichao Zhao, Kepeng Wu, Hezhen Hu, and Houqiang Li. Uni-sign: Toward unified sign language understanding at scale. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2501.15187.

Medet Mukushev, Aigerim Kydyrbekova, Vadim Kimmelman, and Anara Sandygulova. Towards large vocabulary Kazakh-Russian Sign Language dataset: KRSL-OnlineSchool. In *Proceedings of the LREC 2022 Workshop on the Representation and Processing of Sign Languages*, 2022.

Mathias Müller, Zifan Jiang, Amit Moryossef, Annette Rios, and Sarah Ebling. Considerations for meaningful sign language machine translation based on glosses. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

Maja Popović. chrF: Character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation (WMT)*, 2015.

Matt Post. A call for clarity in reporting BLEU

scores. In *Proceedings of the Third Conference on Machine Translation (WMT)*, 2018.

Qwen Team. Qwen3.6-27B, 2026. Model card, https://huggingface.co/Qwen/Qwen3.6-27B.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

Dave Uthus, Garrett Tanzer, and Manfred Georg. YouTube-ASL: A large-scale, open-domain American Sign Language-English parallel corpus. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

Elena Voita, Rico Sennrich, and Ivan Titov. Analyzing the source and target contributions to predictions in neural machine translation. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021.

Ryan Wong, Necati Cihan Camgoz, and Richard Bowden. Sign2GPT: Leveraging large language models for gloss-free sign language translation. In *International Conference on Learning Representations (ICLR)*, 2024.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of NAACL-HLT*, 2021.

Sijie Yan, Yuanjun Xiong, and Dahua Lin. Spatial temporal graph convolutional networks for skeleton-based action recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018.

Benjia Zhou, Zhigang Chen, Albert Clapés, Jun Wan, Yanyan Liang, Sergio Escalera, Zhen Lei, and Du Zhang. Gloss-free sign language translation: Improving from visual-language pretraining. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Hao Zhou, Wengang Zhou, Weizhen Qi, Junfu Pu, and Houqiang Li. Improving sign language translation with monolingual data by sign back-translation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

A Pipeline faults that would have changed a conclusion

Each of these ran without an error, and each would have produced a plausible but wrong result. They are listed in the order we found them.

1. **Validation never saw two sources.** Generation-based model selection used the first 25 unshuffled batches, and because sources were concatenated in order, all 200 clips came from INFORMBURO. QAZAQSTANTV, 48.6% of dev, was never generated on. Replaced by the stratified selection set.
2. **Splits leaked across dataset versions.** Each manifest was internally disjoint, but the archived QAZAQSTANTV train shared 563 videos with the recollected dev (64% of dev clips). This invalidated an earlier clean-versus-noisy data experiment.
3. **Two BLEUs.** The trainer reported normalised BLEU and the evaluator raw BLEU. On identical predictions they gave 63.4 and 30.3, so every BLEU quoted before the fix was normalised and not comparable to published numbers.
4. **Padding leaked into real frames.** Masking only around the temporal ST-GCN stack let convolution bias and BatchNorm shift reach the last real frames (maximum difference 0.65). Fixed by re-zeroing after each block.
5. **Wrists were elbows.** The body mapping copied elbow values into the wrist slots.
6. **The confidence channel was constant.** RTMW scores are ≥ 1 for 99.6–100% of joints, so clipping to $[0, 1]$ produced a channel with standard deviation 0. Upstream Uni-Sign clips the same way.
7. **New heads in no optimiser group.** The alignment heads never trained, and the loss sat at chance.
8. **A guard that consumed its input.** A check that every parameter had an optimiser group iterated over generator-valued groups and emptied them, so mT5 stopped training. Caught by comparing the loss curve with the baseline's.
9. **Wrong sentinel IDs.** On our tokenizer version, $\mathrm{convert_tokens_to_ids}(\text{'<extra_id_0>'})$ returns $<\mathrm{unk}>$, which would have broken span corruption silently.
10. **A mislabelled frame rate.** "Every second frame of a 50 fps video" was true for only one source (Table 1). The E9 smoke test missed this because it built the dataset directly instead of through the trainer's config path.
11. **A one-sample noise floor.** A first parity test of a rewritten trainer declared a bug at 0.79 loss difference against an old-versus-old gap of 0.13. A second run landed 0.30 on the other side. Deterministic mode then gave exact parity over 400 batches.

B LLM rewrite prompt and guard

The prompt is given verbatim (version 2, frozen after the dev pilot). $\{\mathrm{article}\}$ is the $\mathrm{text_human}$ of the clip and its two neighbours on each side, in time order, with consecutive duplicates removed. $\{\mathrm{speech}\}$ is the clip's $\mathrm{text_asr}$.

You are cleaning the reference transcripts of a Kazakh sign-language news corpus.

SPEECH is an automatic speech recognition of exactly what the news anchor said during one short video clip. Its first and last words are the true boundaries of the clip, but words can be misrecognised, and it has no capital letters or punctuation.

ARTICLE is the human-written news text for this broadcast, around the clip. Its wording, spelling, names and numbers are correct, but it is NOT aligned to the clip: it contains sentences said before and after the clip, and it may lack some things that were said.

Write the transcript of exactly what was said in this clip, in clean written Kazakh:
- Cover the same content as SPEECH, from its first word to its last. Do not add anything SPEECH does not contain, and do not drop anything it does.
- Where SPEECH matches the ARTICLE, use the ARTICLE's wording, spelling, capitalisation, numbers and punctuation.
- Where SPEECH has no counterpart in the ARTICLE, keep it, fixing only obvious recognition errors.
- If SPEECH starts or ends in the middle of a sentence, start or end at the same point.
- Speech recognition sometimes repeats a passage by mistake. If SPEECH contains the same passage twice in a row and the ARTICLE has it once, write it once.
- Do not add ellipses or any marks that are not part of the text.

Output only the transcript, nothing else.

ARTICLE:
{article}

SPEECH:
{speech}

Guard. Let a be the ASR with any repeated passage of at least 5 words collapsed, and r the rewrite with ellipses removed. Accept if $0.67 \le |r|/|a| \le 1.5$ in words and $\mathrm{chrF}(r, a) \ge 45$, both computed on lower-cased text with punctuation stripped. On the 600-clip pilot, rejections were 1.8% for length, 0.5% for overlap and 0.8% for a missing ASR or article. Accepted rewrites score chrF 91.8 against the ASR, compared with 72.7 for the original references.

C All from-scratch runs

Run	chrF	recall	INFORMBURO	KHABAR	QAZAQSTANTV	distinct	Δ_{pose} [95% CI]	takeoff
B_s0	29.79	0.128	27.0	33.8	28.2	598	9.52 [8.77, 10.29]	yes
B_s1	21.02	0.031	18.5	24.2	20.0	503	1.27 [0.66, 1.86]	no
B_s2	19.83	0.025	18.6	21.5	19.3	523	1.11 [0.68, 1.56]	no
E4_s0	21.10	0.023	21.2	22.9	18.8	525	0.74 [0.32, 1.17]	no
E4_s1	21.79	0.023	22.6	22.7	19.7	558	0.50 [0.10, 0.94]	no
E4_s2	20.02	0.019	21.1	21.0	17.6	505	0.33 [-0.08, 0.77]	no
E4b_s0	29.73	0.125	27.4	33.2	28.3	600	8.91 [8.22, 9.60]	yes
E4b_s1	31.47	0.145	29.2	35.2	29.6	599	10.68 [9.90, 11.50]	yes
E4b_s2	11.14	0.010	8.1	11.2	14.4	128	0.24 [-0.23, 0.73]	no
E4b_s3	17.17	0.021	14.0	20.1	17.2	300	0.54 [-0.02, 1.09]	no
E4b_s4	18.56	0.019	15.6	20.7	19.3	402	0.63 [0.08, 1.17]	no
E4b_s5	24.24	0.059	22.3	26.8	23.4	597	3.74 [3.19, 4.32]	no
E5_s0	21.70	0.029	20.3	23.9	20.7	566	1.26 [0.76, 1.79]	no
E5_s1	22.10	0.037	20.2	24.7	21.1	586	2.01 [1.50, 2.56]	no
E5_s2	20.95	0.025	18.3	23.7	20.6	521	1.20 [0.69, 1.72]	no
E6_s0	21.39	0.034	19.0	24.3	20.6	567	1.72 [1.20, 2.28]	no
E6_s1	29.31	0.123	25.6	33.5	28.7	598	9.00 [8.24, 9.72]	yes
E6_s2	20.59	0.027	18.6	23.2	19.7	522	1.10 [0.62, 1.57]	no
E7_s0	18.85	0.018	13.5	22.5	20.2	283	0.28 [-0.27, 0.82]	no
E9_s0	17.29	0.015	13.9	19.2	18.7	334	0.50 [-0.02, 1.07]	no
E9_s1	20.18	0.021	19.2	22.4	18.6	437	0.60 [0.14, 1.04]	no
E9b_s0	18.96	0.025	17.4	20.5	18.9	427	1.00 [0.42, 1.55]	no
E9b_s1	19.39	0.025	16.9	21.9	19.2	490	1.18 [0.65, 1.70]	no
E9b_s2	34.57	0.181	31.8	37.8	34.0	599	13.59 [12.75, 14.46]	yes
E9b_s3	27.74	0.106	25.4	31.2	26.3	599	7.66 [6.95, 8.40]	yes
E9b_s4	20.78	0.025	19.9	22.5	19.7	528	0.92 [0.41, 1.42]	no
E9b_s5	19.65	0.018	18.2	22.2	18.2	394	0.49 [-0.02, 1.03]	no
E10_s0	14.42	0.012	16.2	11.9	15.1	133	0.13 [-0.34, 0.62]	no
E10_s1	11.23	0.010	8.8	11.7	13.4	107	0.40 [-0.04, 0.83]	no
E10_s2	17.68	0.014	13.8	19.2	20.0	231	0.35 [-0.20, 0.88]	no
E11_s0	17.05	0.015	9.8	21.4	19.5	250	0.65 [0.21, 1.11]	no
E11_s1	16.55	0.016	11.2	19.3	18.9	227	0.62 [0.10, 1.14]	no
E11_s2	26.59	0.091	24.9	29.5	25.1	598	6.27 [5.62, 6.92]	yes
E13_s0	24.96	0.068	22.5	27.8	24.4	595	4.83 [4.25, 5.41]	no
E13_s1	24.42	0.061	22.6	26.8	23.7	588	3.85 [3.31, 4.41]	no
E13_s2	19.36	0.019	15.6	22.0	20.3	347	0.26 [-0.26, 0.76]	no
E16_s0	35.57	0.191	32.3	38.4	36.0	599	14.26 [13.40, 15.17]	yes
E16_s1	22.70	0.034	20.8	24.1	23.2	579	1.50 [1.03, 1.98]	no
E16_s2	34.76	0.183	30.8	37.6	35.9	598	13.55 [12.71, 14.44]	yes
E16_s3	32.92	0.160	28.7	36.1	34.0	598	11.79 [11.03, 12.59]	yes
E16_s4	22.77	0.036	20.6	24.6	23.0	586	1.74 [1.26, 2.24]	no
E16_s5	17.20	0.017	12.0	20.1	19.3	320	0.34 [-0.26, 0.92]	no

Table 7: Every from-scratch run at epoch 10 on the dev selection set (600 clips, original references), scored through one metric path. "Distinct" is the number of different hypotheses among 600. Contamination (evaluation videos seen in training) is 0 for every run.

D Decoding sweep

The sweep used a single checkpoint on the dev selection set and crossed beam $\{1, 4\}$, repetition penalty $\{1.0, 1.1, 1.3\}$ and trigram blocking on/off. With beam 1 all six settings scored 34.69–34.96 chrF; with beam 4, 36.00–36.27. Beam 4 is consistently better, and the penalty and blocking settings make no difference. The checkpoint for this sweep came from an initialiser excluded from the paper (Appendix E), so only the relative differences are used.

E Excluded results

An mT5 checkpoint fine-tuned by a colleague on an earlier version of this corpus reached 36.25 chrF and 0.207 recall on the clean subset of the selection set. We exclude it because its training history is unrecorded and may overlap our dev and test videos. All results in the paper start from public weights (Uni-Sign CSL stage 1 and $\mathrm{google/mt5\text{-}base}$). Earlier prosody and speech-synthesis stages, and an RGB fusion branch, were dropped for scope; their code is in the project history.

F Compute

Each from-scratch run took about 5.7 h, and a fine-tune about 1.5 h, on one NVIDIA RTX PRO 6000 (96 GB). The 42 from-scratch runs account for roughly 240 GPU-hours. Scoring a checkpoint on the selection set takes about 40 s, plus the same again for the shuffled pass, and the LLM rewrite of 34,768 clips took 72 minutes on two GPUs.